YEAR 7 · UNIT 4 REVIEW · SCIENCE UNIT 3 · HOMEWORK 10

# Faster in January

Part 1: Maths — Unit 4 review
Part 2: Science — 3.2, 3.3 and the football model, with decimals
Part 3: Reading graphs with decimals

**Name:** _____ **Class:** _____

**Date given:** _____ **Date due:** Friday 30 October 2026

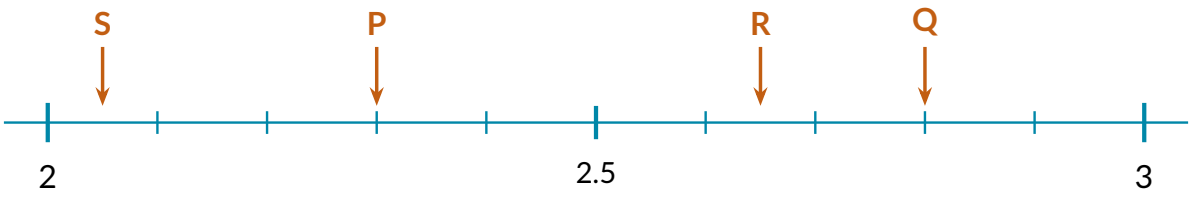
**How to do this homework**

- **Sections A–C:** maths (Unit 4). **D–F:** science, and F uses decimals. **G:** reading graphs.
- No calculator. Estimate first. Then write the calculation, then the answer.
- Answer in full sentences when it says so. *"Gravity"* is not a sentence. *"Gravity pulls the planet towards the Sun"* is.

## Section A · Unit 4 Review — Comparing, Adding and Subtracting

### A1 Read the line

The line goes from **2** to **3**. Every small line is **0.1**. **(a)** Write the number each arrow points to.




| | | | |
|---|---|---|---|
| P _____ | Q _____ | R _____ | S _____ |

**(b)** Write the four numbers in order. Start with the **largest**.

_____

### A2 Which is bigger?

Write <, > or = in each box.

*Tip:* compare the whole numbers first. Then add zeros: 0.7 = 0.70, so 0.7 > 0.65.

(a) 0.5 ☐ 0.45

(b) 2.1 ☐ 2.09

(c) 8.08 ☐ 8.8

(d) 6.4 ☐ 6.400

(e) 15.6 ☐ 9.87

(f) 1.25 ☐ 1.3

## A3 Line up the points

**Written method**

Write the numbers with the **points in a line**. Fill the gaps with **zeros**. The point in the answer goes **straight down**.

**(a)** 7.8 + 4.56

**(b)** 23.4 + 0.87 + 5

**(c)** 16.25 + 3.9

**(d)** 10 − 3.42

**(e)** 6.3 − 2.75

**(f)** 0.9 − 0.36

## Section B · Unit 4 Review — Multiplying and Dividing

**Where the point goes**

**Multiply** Take the point out and multiply. Then **estimate** to put it back.
*Example:* 3.8 × 12 → 38 × 12 = 456 → estimate 4 × 12 = 48 → **45.6**

**Divide** Short division. The point goes **straight up**. If a digit is too small to divide, write **0**. Dividing by a 2-digit number? Write its **times table** first.

## B1 Point out, point back

Fill in the table. Row 1 is done for you.

| Calculation | Estimate | Multiply without the point | Answer |
|---|---|---|---|
| 2.4 × 13 | 2 × 13 = 26 | 24 × 13 = 312 | 31.2 |
| 0.6 × 9 | | | |
| 3.45 × 6 | | | |
| 7.2 × 15 | | | |
| 2.8 × 23 | | | |

## B2 The point goes straight up

Use short division in each box. For **(e)** and **(f)**, write the 15 and the 24 times tables first.

**(a)** 9.45 ÷ 5

**(b)** 4.32 ÷ 4

**(c)** 0.72 ÷ 9

**(d)** 7.3 ÷ 2

**(e)** 52.5 ÷ 15

**(f)** 81.6 ÷ 24

## Section C · Unit 4 Review — Words, Quick Ways and Money

## C1 From English to a calculation

**Word help — the words that choose the operation**

| | |
|---|---|
| **each** | for one of them. *3 pens at 2 dollars each*: 3 × 2 |
| **altogether** | all of it, the total |
| **twice as far** | 2 times as far |
| **What must you add to 4 to make 7?** | 7 − 4. It says *add*, but you **subtract** |
| **3 less than 10** | 10 − 3. The **second** number comes **first** |

Write the **calculation first**. Then write the answer, with its unit if it has one (m, km).

(a) What must you add to 4.65 to make 7?

**Calculation:** _____ **Answer:** _____

(b) A ribbon 2.4 m long is cut into 16 equal pieces. How long is each piece?

**Calculation:** _____ **Answer:** _____

(c) Mr Bowen walks 2.5 km on Monday. On Tuesday he walks twice as far. How far does he walk altogether?

**Calculation:** _____ **Answer:** _____

(d) What is 3.5 less than 10?

**Calculation:** _____ **Answer:** _____

## C2 Now you write the question

Write a word problem for $12 \times 2.40 = 28.80$

**Your word problem must have: tick each one**

- ☐ a real situation
- ☐ a word from C1
- ☐ full sentences
- ☐ units (dollars, kg)
- ☐ a question at the end
- ☐ the answer 28.80

## C3 Use a quick way

Use the **same quick way** as the example. Write the quick way, then the answer.

| Quick way | Example | Your turn |
|---|---|---|
| × 9: × 10, then take away one lot | $3.6 \times 9 = 36 - 3.6 = 32.4$ | (a) $4.7 \times 9 =$ _____ = _____ |
| × 11: × 10, then add one lot | $2.3 \times 11 = 23 + 2.3 = 25.3$ | (b) $6.2 \times 11 =$ _____ = _____ |
| **Split** the number | $4.3 \times 6 = 24 + 1.8 = 25.8$ | (c) $5.3 \times 4 =$ _____ = _____ |
| × 0.25 = a quarter of | $28 \times 0.25 = 28 \div 4 = 7$ | (d) $64 \times 0.25 =$ _____ = _____ |
| × 0.07 = ÷ 100, then × 7 | $200 \times 0.07 = 2 \times 7 = 14$ | (e) $300 \times 0.07 =$ _____ = _____ |
| × 0.5 = half of | $7.4 \times 0.5 = 7.4 \div 2 = 3.7$ | (f) $9.6 \times 0.5 =$ _____ = _____ |

## C4 Change from 10 dollars

Money has **2 decimal places**: 7.8 dollars is written 7.80 dollars.

Mr Bowen buys 3 bánh mì at 1.35 dollars each and 2 iced coffees at 1.90 dollars each. He pays with a 10-dollar note. How much change does he get?

**1.** 3 bánh mì: _____ dollars

**2.** 2 iced coffees: _____ dollars

**3.** Total: _____ dollars

**4.** Change: _____ dollars

Working

## C5 Mr Bowen's homework

Only **one** of Mr Bowen's answers is right. Write **right** or **wrong** for each one. If it is wrong, write the correct answer.

| Mr Bowen wrote | Right or wrong? | Correct answer |
|---|---|---|
| (a) 4.5 + 2.7 = 6.12 | | |
| (b) 9.6 ÷ 3 = 3.2 | | |
| (c) 3.40 > 3.4 | | |
| (d) 60 × 0.4 = 2.4 | | |

# Section D · Science 3.2 — How the Solar System Formed

## D1 Key words

Match each word to its meaning. Write the letter (A–F) in the box.

**nebula** ☐
**hypothesis** ☐
**evidence** ☐
**model** ☐
**limitation** ☐
**plane** ☐

**A** an idea that can be tested
**B** one flat surface
**C** a cloud of dust and gas in space
**D** facts from observations or experiments
**E** something a model cannot do, gets wrong or leaves out
**F** a way of showing something that is hard to observe directly

## D2 Support or contradict?

The **hypothesis**: **The Sun and the planets formed from one spinning cloud of dust and gas.**

Does each fact **support** it (agree) or **contradict** it (go against it)? Tick **one** box.

| Fact | supports | contradicts |
|---|---|---|
| All the planets orbit the Sun in the same direction. | | |
| All the orbits are in the same plane. | | |
| Venus spins the opposite way to most planets. | | |
| HL Tauri is a young star with a flat disc of dust round it. | | |
| Uranus spins on its side. | | |
| Most moons orbit their planet in the same direction. | | |

**(b)** Two facts contradict the hypothesis. Is the hypothesis wrong? Answer in a full sentence. Use the words **most of the evidence**.

## D3 From a cloud to the Sun and planets

**(a)** Put the four steps in order. Write **1**, **2**, **3** and **4** in the boxes.

☐ The hot middle becomes the Sun.

☐ Gravity pulls the nebula together. It spins faster.

☐ Dust sticks together and grows into planets.

☐ The nebula flattens into a spinning disc.

**(b)** Why do all the planets orbit the Sun in the same direction? Answer in a full sentence. Start: **"They all formed from …"**

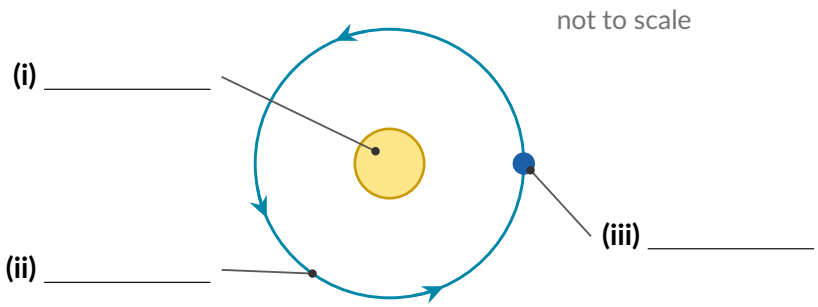
**(c)** Mercury, Venus, the Earth and Mars are small and rocky. Why? Start: **"Near the Sun, it was too hot for …"**

**(d)** The Sun has **99.8%** of the Solar System's mass. What % is left for the rest? _____ %

# Section E · Science 3.3 — Movement in Space

## E1 What keeps a planet in orbit?

**(a)** Label the drawing. Use these words: **Sun** · **orbit** · **planet**




(b) Draw an **arrow** on the planet. Show which way the Sun's **gravity** pulls it.

(c) The arrows show which way the planet moves. Now the Sun's gravity **switches off**. Draw a **dashed line** to show where the planet goes.

## E2 Which forces?

Which forces act on each one? Tick **every** box that is true. Some rows need **two** ticks.

| | gravity | air resistance |
|---|---|---|
| The Earth, orbiting the Sun | | |
| The Moon, orbiting the Earth | | |
| A ball falling from the balcony | | |
| A plane landing on a runway | | |

**(b)** Why do the planets never slow down? Start: **"Space is a …, so …"**

# Section F · Science in Decimals — Faster Near the Sun

**Speed, from Science 3.3**

**Speed**: how fast something moves. **km/s** = kilometres **per** second.
As the distance from the Sun **increases**, the speed of the planets **decreases**. The Sun's pull is **weaker** far away.

## F1 How much faster? How many times as fast?

| | Mercury | Venus | Earth | Mars | Jupiter | Saturn | Uranus | Neptune |
|---|---|---|---|---|---|---|---|---|
| **speed** (km/s) | **48** | **35** | **30** | **24** | **13** | **10** | **7** | **5** |

Fill in the table. Row 1 is done for you. In each row, the **first** planet is the faster one.

| **Planets** | **How much faster?** **subtract**: the difference | **How many times as fast?** **divide**: big ÷ small |
|---|---|---|
| Mercury and Neptune | 48 − 5 = 43 km/s | 48 ÷ 5 = 9.6 |
| Mercury and the Earth | | |
| Mars and Saturn | | |
| The Earth and Mars | | |

## F2 The Sun is a football

In the lesson the Sun became a football, and everything else shrank by the same amount.

| | the Moon | the Earth | Neptune | Jupiter | the Sun |
|---|---|---|---|---|---|
| **Model width** | 0.5 mm | 2 mm | 7.8 mm | 2.3 cm | 22 cm |
| **Like** | a grain of sand | a pinhead | a pea | a grape | a football |

**(a)** Write the five widths in order, **smallest** first. Remember: **1 cm = 10 mm**.

_____

(b) Mr Bowen puts the football against one wall of his classroom, which is **8.5 m** long. The Earth must be **24 m** from the football. How far past the other wall is the Earth?




**Calculation:** _____ **Answer:** _____

(c) Give **one limitation** of the model. Start: **"One limitation is …"**

_____

_____

## F3 Faster in January

The Earth's orbit is *almost* circular. In January the Earth is a little **closer** to the Sun than in July.

| | Distance from the Sun | Speed |
|---|---|---|
| **January** | 147.1 million km | 30.3 km/s |
| **July** | 152.1 million km | 29.3 km/s |

(a) How much closer to the Sun is the Earth in January?

**Calculation:** _____ **Answer:** _____

(b) In which month is the Earth faster? Why? Use the words **closer** and **pull**. Start: **"The Earth is faster in …, because …"**

_____

_____

(c) How much faster is the Earth in January than in July?

**Calculation:** _____ **Answer:** _____

(d) Find the **mean** of the two speeds.

**Calculation:** _____ **Answer:** _____

(e) Round your mean to the nearest whole number. _____

(f) In the lesson, the Earth's speed was **30 km/s**. What do you notice?

_____

_____

## Section G · Reading Graphs with Decimals

### Reading a scale

1. Find two **numbers** next to each other on the scale.
2. Count the **gaps** between them — not the ticks.
3. One tick = the difference ÷ the number of gaps.



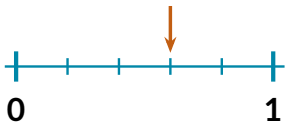

5 gaps: 2 ÷ 5 = 0.4. One tick is **0.4**.

**Word help:** the **axis** is the line along the bottom or side · the **scale** is its numbers · a **tick** is a small line on it · **flat** = not going up or down · **steep** = going up fast

### G1 What is one tick worth?

Write what **one tick** is worth. Then write the number the arrow points to.

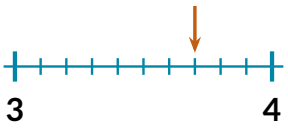
**(a)**




1 tick: _____

Arrow: _____

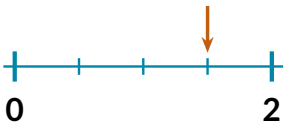
**(b)**




1 tick: _____

Arrow: _____

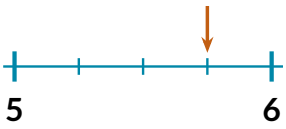
**(c)**




1 tick: _____

Arrow: _____

**(d)**

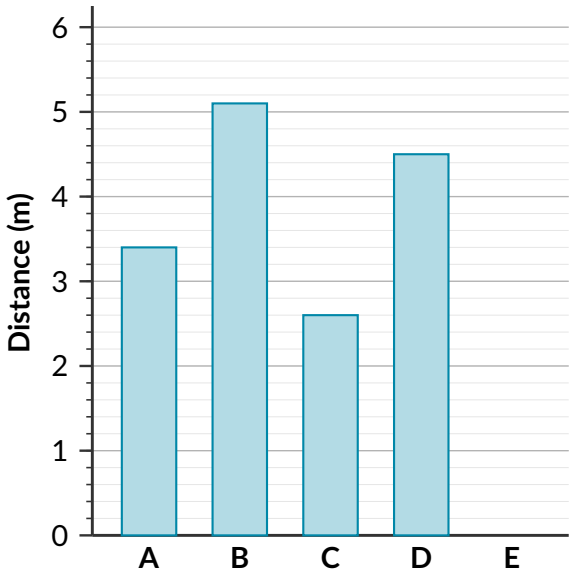



1 tick: _____

Arrow: _____

### G2 Mr Bowen's paper planes

In a long meeting, Mr Bowen made four paper planes. The bar chart shows how far each one flew.




**(a)** One small tick = _____ m

**(b)** How far did each plane fly?

A _____ m B _____ m

C _____ m D _____ m

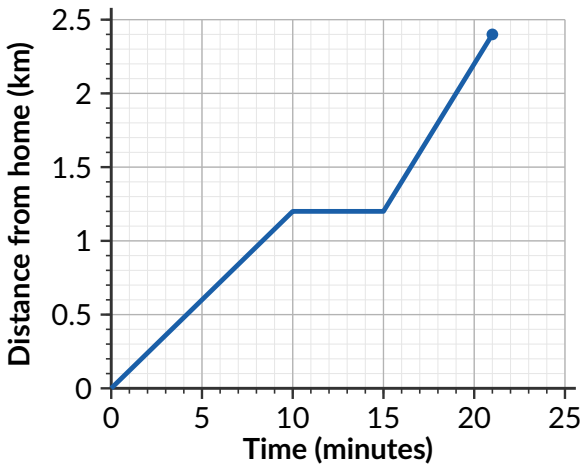
**(c)** How much further did the best plane fly than the worst one?

_____ = _____ m

**(d)** Plane E flew **3.9 m**. Draw its bar.

## G3 Mr Bowen walks to school

Mr Bowen walks from home to school. On the way, he stops at a café. The graph shows how far he is from home.




**(a)** One small tick:
distance _____ km, time _____ min

**(b)** How far is the café from home? _____ km

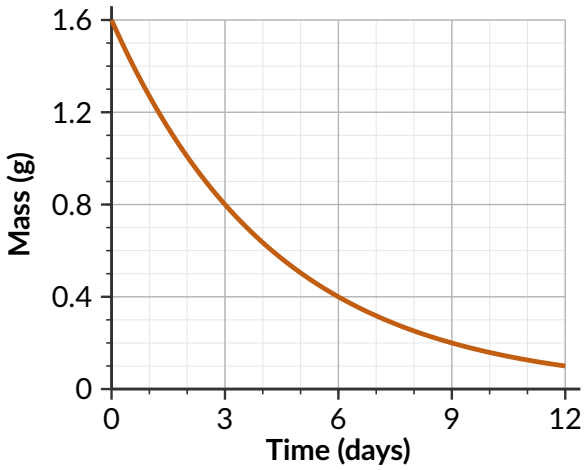
**(c)** How long is he at the café? _____ min

**(d)** How far is school from home? _____ km

**(e)** How far from home is he after **18 minutes**?
_____ km

## G4 A half-life graph

A scientist weighs a sample of a metal that breaks down, like carbon-14 but much faster. The graph shows its mass.




**(a)** One small tick on the mass axis = _____ g

**(b)** The mass at the start: _____ g

**(c)** The mass is 0.8 g after _____ days. So the **half-life** is _____ days.

**(d)** After 6 days: _____ g. After 9: _____ g.

**(e)** After 2 days: about _____ g.

**(f)** Mr Bowen says: *"After 12 days it is all gone."*
After 12 days there is _____ g left.

So he is ☐ right ☐ wrong.

## G5 Tell the story of the graph

Write the story of the **G3** graph in **4 or 5 full sentences**. Tick each one:

☐ three numbers, with units (km, minutes)

☐ what he does at the café, and how long

☐ **faster** before or after the café? How do you know?

☐ a reason he might be faster

## Answer Key — Teacher Copy

### A1 Read the line

(a) P 2.3 · Q 2.8 · R 2.65 · S 2.05
(b) 2.8, 2.65, 2.3, 2.05

R and S sit halfway between two lines; HW 9's A1 asked the class to *place* 5.45 and 5.05, and this asks them to *read* them. Expect 2.5 for S from a student who sees "halfway" and writes a 5 in the tenths, and 2.6 or 2.7 for R. (b) runs from the *largest* on purpose — HW 9 always ran from the smallest. A list that starts with 2.05 is right in the wrong direction: ask what *largest* means.

### A2 Which is bigger?

(a) > (b) > (c) < (d) = (e) > (f) <

(a) and (f) are the same trap from two sides: the number with more digits is the smaller one (0.50 > 0.45, 1.25 < 1.30). (c) is "eight point zero eight" against "eight point eight"; anyone who reads both as "eight point eight" has dropped the zero. (e) is the whole-number part first: 15 > 9, and the digits after the point do not matter.

### A3 Line up the points

(a) 12.36 (b) 29.27 (c) 20.15 (d) 6.58 (e) 3.55 (f) 0.54

(a) and (c) are the alignment check: lining up the right-hand digits gives 7.8 + 4.56 = 5.34 and 16.25 + 3.9 = 16.64. (b) has a whole number with no point: 5 = 5.00. (d), (e) and (f) need the zeros *written* first: 10.00, 6.30, 0.90. Expect 7.42 for (d) from a student who took 10 − 3 and copied the .42.

### B1 Point out, point back

| | | | |
|---|---|---|---|
| $0.6 \times 9$ | $1 \times 9 = 9$ | $6 \times 9 = 54$ | **5.4** |
| $3.45 \times 6$ | $3 \times 6 = 18$ | $345 \times 6 = 2070$ | **20.7** |
| $7.2 \times 15$ | $7 \times 15 = 105$ | $72 \times 15 = 1080$ | **108** |
| $2.8 \times 23$ | $3 \times 20 = 60$ | $28 \times 23 = 644$ | **64.4** |

Rows 2 and 3 end in a zero once the point is back (20.70, 108.0); accept them with or without it. Row 3 is the one to watch: $72 \times 15 = 1080$, and a student who puts back a place for *each* number writes 10.80. Only 7.2 has a decimal place. The estimate, 105, settles it.
Row 1 is $0.6 \times 9$, not $6 \times 9$: expect 0.54 from anyone counting all the digits instead of the decimal places.
Any sensible estimate is right ($2.8 \times 23 \approx 3 \times 23 = 69$ is fine).

### B2 The point goes straight up

(a) 1.89 (b) **1.08** (c) **0.08** (d) **3.65** (e) 3.5 (f) 3.4

(b) needs one 0 (4 into 3 will not go); expect 1.8. (c) needs two: 9 into 0, then 9 into 7, will not go. Expect 0.8, which is $0.72 \div 0.9$.
(d) runs out of digits: write 7.30 and keep going. "3.6 remainder 1" is the usual stop.
(e) and (f): the table is the method. $15 \times 3 = 45$, 7.5 left = $15 \times 0.5$. $24 \times 3 = 72$, 9.6 left = $24 \times 0.4$. A student with the right answer and no table probably knew it; one with a wrong answer and no table needs telling to write the table.

## C1 From English to a calculation

(a) 7 − 4.65 = 2.35 (b) 2.4 ÷ 16 = 0.15 m
(c) 2.5 × 2 = 5, 2.5 + 5 = **7.5** km (d) 10 − 3.5 = 6.5

**Mark the calculation, not the answer.** This is the English question.
(a) says *add* and is a subtraction. 4.65 + 7 = 11.65 is the English error, not an arithmetic one: have the student read the word-help line aloud.
(d) is "3.5 less than 10": 3.5 − 10 is reading left to right.
(c) is two steps and *altogether* is the second. Expect 5 km, which is Tuesday alone.
(b) is the 16 table: 16 × 0.1 = 1.6, 0.8 left = 16 × 0.05. Expect 1.5 from a student who did 24 ÷ 16 and put the point back in the wrong place; the estimate (2.4 m in 16 pieces is much less than 1 m each) catches it.

## C2 Now you write the question

Check, in order: a real situation; full sentences; ends in a question mark; uses a word from C1; gives units; and it really comes out at 28.80.

A good answer: "Mr Bowen buys 12 notebooks at 2.40 dollars each. How much does he pay altogether?"
The common failure is a question whose answer is 0.20: "12 friends share 2.40 dollars". Read it back to them and ask what the calculation in the box does.
Units need not be dollars — "12 bags of rice, 2.40 kg each" is just as good — but 28.80 with two decimal places is a strong hint.

## C3 Use a quick way

(a) 47 − 4.7 = 42.3 (b) 62 + 6.2 = 68.2 (c) 20 + 1.2 = 21.2
(d) a quarter of 64 = 16 (e) 3 × 7 = 21 (f) half of 9.6 = 4.8

Each row's example uses the same quick way as its question, so a student who copies the example's shape gets there. (a) needs 47.0 − 4.7. Expect 43.7 from a student who took 47 − 4 and copied the .7 across — the A3(d) error again.
(e) divides by **100**, not 10. Expect 210 (30 × 7).
Accept a correct written method for any of them, but the column is "the quick way": ask for the quick way.

## C4 Change from 10 dollars

3 × 1.35 = 4.05, 2 × 1.90 = 3.80, 4.05 + 3.80 = 7.85, 10.00 − 7.85 = **2.15** dollars

Four steps. The usual slip is to stop at 7.85 and call it the change; ask what the question asked for. 3.8 for the coffees is right but is money, so 3.80.
10.00 − 7.85 needs the zeros written. Expect 3.85 from a student who took 10 − 7 = 3 and copied the .85 — A3(d) and C3(a) again.

## C5 Mr Bowen's homework

(a) **wrong** — 7.2. He added 4 + 2 and 5 + 7 separately; the 12 tenths carry.
(b) **right.**
(c) **wrong** — 3.40 = 3.4. The zero at the end does not change anything.
(d) **wrong** — 24. × 0.4 is ÷ 10, then × 4: 6 × 4.

One trap each from 4.2, 4.4, 4.1 and 4.5. Whichever one a student gets wrong, send them back to A3, B2, A2 or C3 rather than explaining again.
(b) is right on purpose and is the one with most room for doubt: a student who expects every division to need a 0 will "correct" it to 3.02.

## D1 Key words

nebula **C** · hypothesis **A** · evidence **D** · model **F** · limitation **E** · plane **B**

These are the 3.2 and Models write notes, word for word. Hypothesis and evidence (A and D) are the pair most often swapped; *plane* is the one most often left blank ("not an aeroplane").

## D2 Support or contradict?

Supports: rows 1, 2, 4, 6. Contradicts: rows 3 (Venus), 5 (Uranus).
(b) "No. Most of the evidence supports the hypothesis, so it is not wrong. It is just not proven."

(b) is the deck's Question 4 ("most, but not all, of the evidence supports it"). Accept any sentence that says the hypothesis is kept because most evidence supports it. A student who says "yes, it is wrong" has heard *contradict* as *prove wrong*; the cake on deck slide 6 is the reminder — two clues still supported the cat.
Row 4 (HL Tauri) is the one students hesitate over. It supports: a young star with a flat disc is the hypothesis happening today.

## D3 From a cloud to the Sun and planets

(a) In the order printed: **3, 1, 4, 2**.
(b) "They all formed from one spinning cloud, so they all still go round the same way."
(c) "Near the Sun, it was too hot for ice, so only rock and metal were left to make planets."
(d) 100 − 99.8 = **0.2**%

(a) The printed order is shuffled; read it as "the hot middle is 3". The usual error is putting the Sun first, which is the order of importance, not of time.
(c) is the Models deck's "Rocky or giant?" slide; "too hot for ice" is the point. Accept "too hot for gas" only if they also say the planets are made of rock and metal.
(d) needs 100.0 − 99.8. Expect 1.8 (100 − 99, then the .8 copied) or 1.2 from anyone who did not write the zero. Every planet, moon, asteroid and comet together is two tenths of one percent: that is worth saying aloud.

## E1 What keeps a planet in orbit?

(a) (i) Sun · (ii) orbit · (iii) planet
(b) An arrow from the planet pointing **to the left**, at the Sun.
(c) A straight dashed line going **straight up the page** from the planet — along the direction it was moving, touching the orbit, not crossing it.

(c) is the class vote from the lesson (left hand "away from the centre", right hand "straight on"), and the wrong answer is the same here: a line straight out to the right, away from the Sun. It means the student thinks something was pushing the planet outwards. Nothing was — gravity was pulling it in, and when gravity stops the planet simply keeps going the way it was going.
A curved dashed line that keeps going round is the other error: without gravity nothing turns it.

## E2 Which forces?

Earth: gravity · Moon: gravity · falling ball: gravity *and* air resistance · plane: gravity *and* air resistance
(b) "Space is a vacuum, so there is no air resistance to slow the planets down."

Row 1 is the deck's Question 6, and row 2 is the same answer for the Moon: gravity only. Expect air resistance ticked for the Earth by students who picture the Earth's own air; it is the planet moving through space that matters, and space has no air.
Rows 3 and 4 are on Earth, where there is air. The plane is the photo from the air resistance slide.
(b): "vacuum" and "no air resistance" are the two things to look for.

## F1 How much faster? How many times as fast?

| | | |
|---|---|---|
| Mercury and the Earth | 48 − 30 = 18 km/s | 48 ÷ 30 = 1.6 |
| Mars and Saturn | 24 − 10 = 14 km/s | 24 ÷ 10 = 2.4 |
| The Earth and Mars | 30 − 24 = 6 km/s | 30 ÷ 24 = 1.25 |

**The two columns are the point.** A student who writes the same kind of answer in both has not read the questions. "How many times" has no unit: 1.6, not 1.6 km/s.
30 ÷ 24 is the review's 2-digit division: 24 × 1 = 24, 6 left, 60 ÷ 24 = 2 remainder 12, 120 ÷ 24 = 5, so 1.25. Expect 1.2 remainder 12.
48 ÷ 30 and 24 ÷ 10 need a 0 written: 48.0, 24.0. Expect 1.6 to come out as "1 remainder 18".

## F2 The Sun is a football

(a) 0.5 mm, 2 mm, 7.8 mm, 2.3 cm, 22 cm (b) 24 − 8.5 = **15.5** m
(c) Any one: nothing moves (the planets do not orbit or spin); the football is not hot and gives no light; most of the planets and moons are missing; a 0.5 mm Moon is too small to see.

(a) is a units trap. A student who ignores the units puts 2.3 before 7.8 — Jupiter smaller than Neptune — and the grape and the pea in the table say otherwise. 2.3 cm is 23 mm.
(b) needs 24.0 − 8.5. The pinhead goes 15.5 m out into the corridor, which is the joke and also the point of the Models lesson: the Solar System is mostly empty.
(c) is the Cambridge question. Any honest limitation in a full sentence is right; "it is not real" is not a limitation — no model is real.

## F3 Faster in January

(a) 152.1 − 147.1 = **5** million km
(b) "The Earth is faster in January, because it is closer to the Sun and the Sun's pull is stronger."
(c) 30.3 − 29.3 = **1** km/s (d) 30.3 + 29.3 = 59.6, 59.6 ÷ 2 = **29.8** km/s
(e) **30** (f) "It is the same as the Earth's speed in the lesson."

**This is the question the packet is built around.** It is 3.3's exit question with real numbers (NASA: 147.1 and 152.1 million km; 30.29 and 29.29 km/s). The rule from 3.3 is that closer planets move faster, and here it holds for one planet over one year.
(f) is the reveal: the book's 30 km/s is the Earth's *mean* speed, rounded. Worth putting on the board: the table in F1 is a table of means. Any sentence that says the two are the same earns (f); one that says *why* ("the lesson used the mean") is the best answer on the packet.
(a) and (c) both come out as whole numbers written as decimals (5.0, 1.0); accept either. (b) is 3.3's "closer planets move faster" turned on one planet: accept any sentence that says January and gives the reason. "Because it is closer" alone is half of it — ask *why* closer means faster, and look for *pull*.
Do not let this become a lesson on seasons. January is not the hottest month, and the reason (the tilt of the Earth) has not been taught. If someone asks, that is a good question to leave open.

## G1 What is one tick worth?

(a) tick **0.2**, arrow **0.6** · (b) **0.1**, **3.7** · (c) **0.5**, **1.5** · (d) **0.25**, **5.75**

**The error to catch: counting ticks, not gaps.** (a) has 4 ticks between 0 and 1 but 5 gaps; a student who divides by 4 writes 0.25 and then 0.75 for the arrow. (c) gives 2 ÷ 3, which will not come out, and (d) gives 0.333 . . . — a tick that "does not work" is the clue. Send them back to the remember box's 0-to-2 strip, where the gaps are numbered. (b) is A1's 0.1 again, the easy one.

## G2 Mr Bowen's paper planes

(a) **0.2** m · (b) A 3.4, B 5.1, C 2.6, D 4.5 m · (c) 5.1 − 2.6 = **2.5** m · (d) a bar halfway between the 3.8 and 4.0 lines

A and C end on a tick; B and D end *halfway* between two, so expect 5.0 or 5.2 for B and 4.4 or 4.6 for D — ask what is halfway between them. A student with 0.5 for (a) has counted the big squares as ticks and will read B as 5.5. (c) is "further": the best minus the worst, not the total (15.6 m).

## G3 Mr Bowen walks to school

(a) **0.1** km, **1** minute · (b) **1.2** km · (c) **5** minutes · (d) **2.4** km · (e) **1.8** km

(c) is the flat part, from 10 to 15 minutes: expect 15 (the time he *leaves*) or 10 (the time he arrives); the question is how long, so subtract. (e) 18 minutes is halfway through the last part: 1.2 km plus half of 1.2. The 0.1 km tick is the same trap as G1(a): 5 gaps between 0 and 0.5.

## G4 A half-life graph

(a) **0.1** g · (b) **1.6** g · (c) **3** days; the half-life is **3** days · (d) **0.4** g; **0.2** g · (e) about **1.0** g (exactly 1.008; accept 0.9–1.1) · (f) **0.1** g; **wrong**

(a) has 4 gaps between labels 0.4 apart: 0.4 ÷ 4 = 0.1. A student who reads 0.4 per small square has used the big squares. (e) is between grid lines, like the deck's bedtime slide: anything near 1.0 shows they followed the curve, not a straight line from 1.6 to 0.8 (which gives 1.07, also fine). (f) is the deck's "Does it ever run out?": the model halves for ever, so there is always some left.

## G5 Tell the story of the graph

Mark the four ticks in order. A model answer:
*Mr Bowen walks 1.2 km from his home to a café. It takes him 10 minutes. He stays at the café for 5 minutes, and the line is flat because he is not moving. Then he walks 1.2 km more to school in only 6 minutes. He is faster after the café, because the line is steeper. Maybe the coffee made him walk faster.*

**The thinking is the steeper line.** Same distance (1.2 km) in less time (6 minutes, not 10), so faster. A student who says "faster before, because the first part is longer" has read the length of the line, not its slope: ask how many minutes each part took. Any sensible reason earns the last tick; the coffee is the one the packet is fishing for (Extra Homework 10). Three numbers *with units* — "1.2" alone does not count.